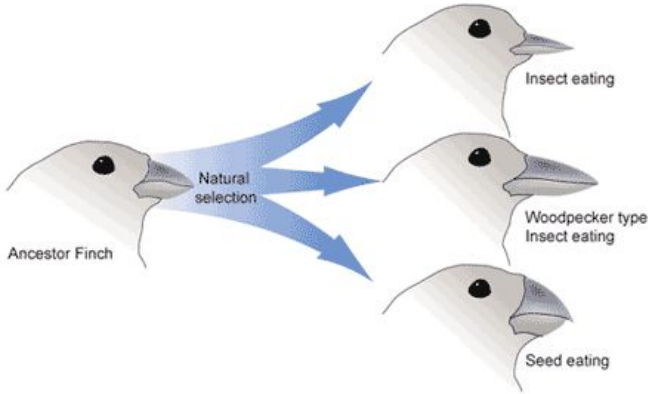
BME 105: Genetics in the Genomics Era

Lecture 21: Natural selection



Instructor: Dr. Shelbi Russell,
Assistant Professor

Course website on canvas.ucsc.edu

Lecture 21 Learning Objectives

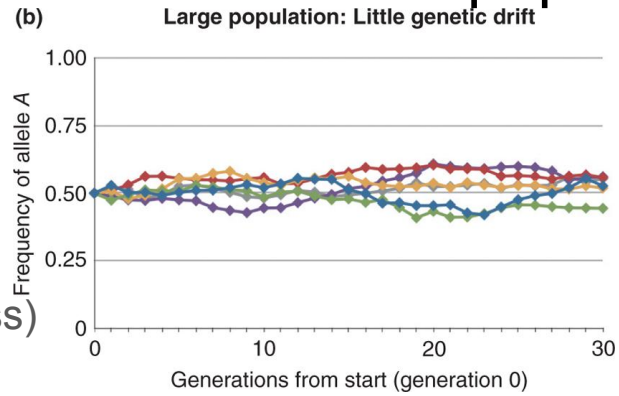
- Be able to describe what "selected sites" and "neutral sites" are in the genome
- Understand how selection acts on advantageous and deleterious alleles
- Understand relative fitness and how HWE values can be rescaled to account for selection
- Be able to calculate *W*, the relative fitness normalization factor
- Understand how selection impacts allele frequency change in heterozygotes
  - exponential decay of q^2 due to heterozygotes hiding recessive alleles from selection
- Be able to distinguish "mutation" from "substitution"
- Be able to describe molecular conservation/constraint due to stabilizing selection
- Understand balancing selection
- Be able to describe how the ratio of non-synonymous to synonymous mutations can be used to detect selection

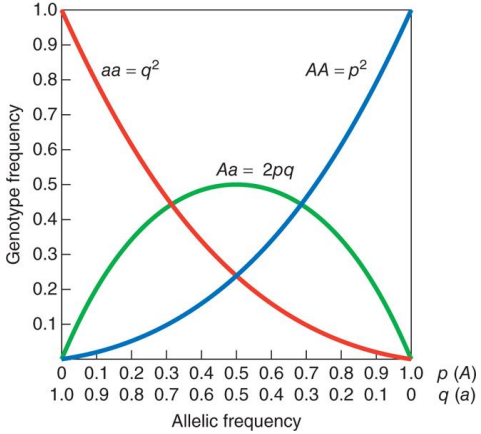
Hardy-Weinberg Equilibrium (HWE) - genetic variation will remain constant across generations in an ideal population

Five simplifying assumptions:

- Random mating
- No mutation
- No gene flow (no migration)
- No selection (no difference in survival or fitness)
- Infinitely large population (no drift)

⇒ no evolution



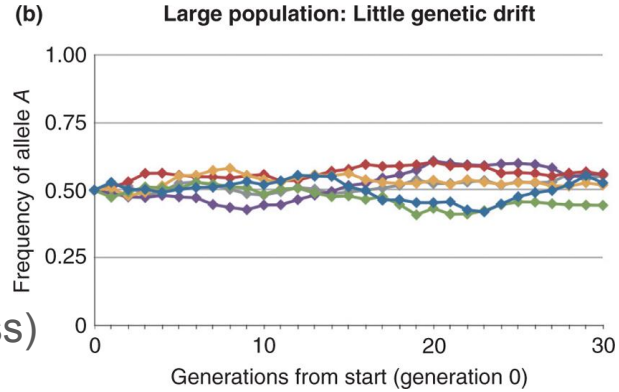


How do you get changes in allele frequencies?

Violate any of the five simplifying assumptions:

- Random mating
- No mutation
- No gene flow (no migration)
- No selection (no difference in survival or fitness)
- Infinitely large population (no drift)

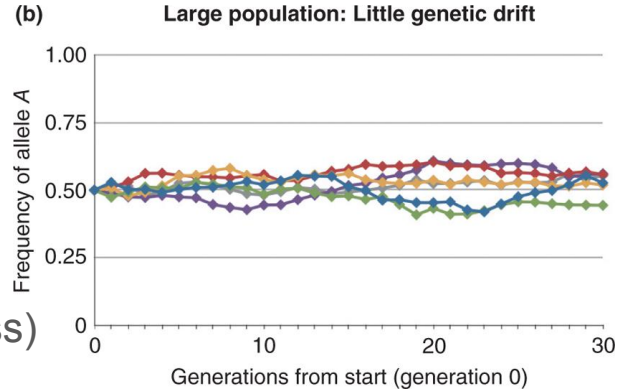
⇒ evolution (of some kind) occurred in the recent past



How do you get changes in allele frequencies?

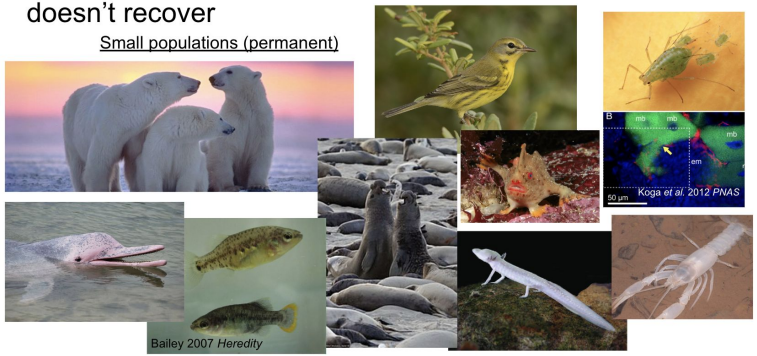
Violate any of the five simplifying assumptions:

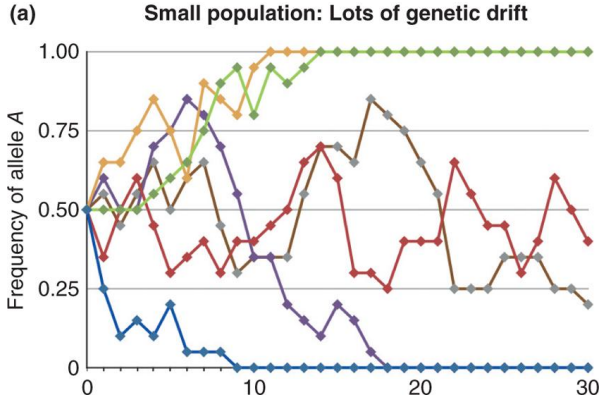
- Random mating
- No mutation
- No gene flow (no migration)
- No selection (no difference in survival or fitness)
- <u>Infinitely large population (no drift)</u>



Genetic drift: a permanent issue if small population size doesn't recover

<u>Small populations (permanent)</u>

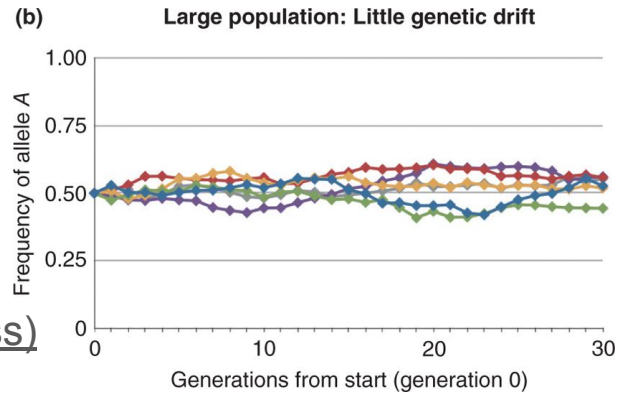


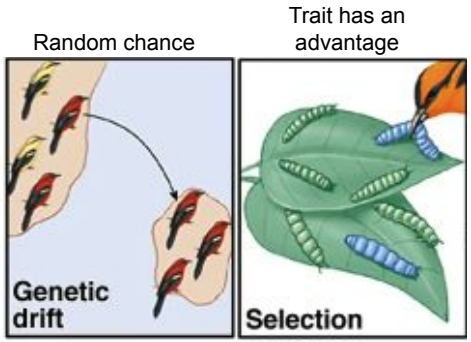


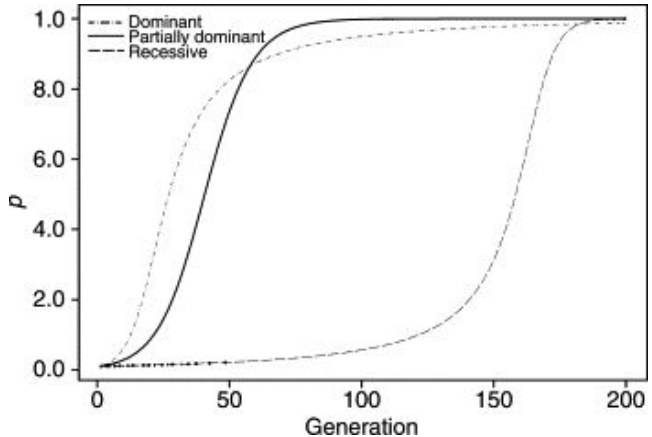
How do you get changes in allele frequencies?

Violate any of the five simplifying assumptions:

- Random mating
- No mutation
- No gene flow (no migration)
- <u>No selection (no difference in survival or fitness)</u>
- Infinitely large population (no drift)

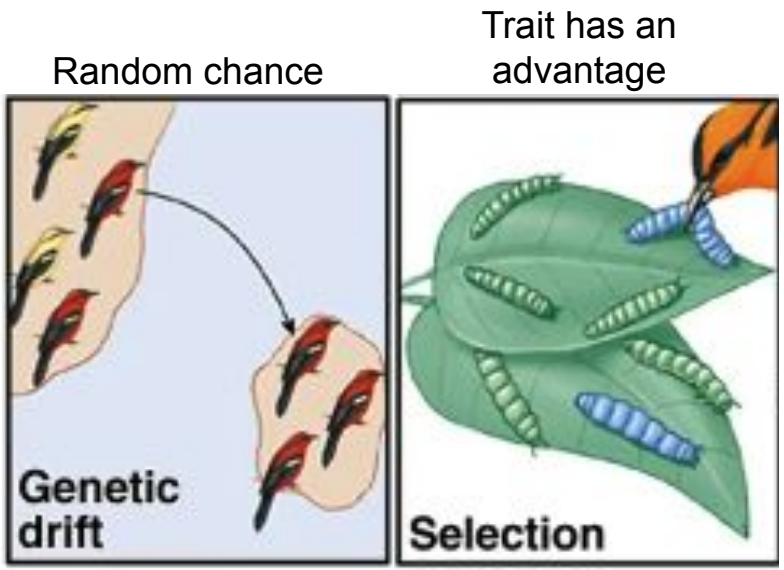






Natural selection alters allele frequencies because they experience higher rates of transmission to offspring

Genetic drift =
Alleles change in frequency due to random chance



Natural selection =
Alleles that determine an advantageous trait increase in frequency if more of them survive or transmit each generation

Alleles that determine deleterious traits decrease in frequency…

Natural selection affects allele frequencies

Natural selection in pocket mice depends on the environment

Reprinted from PNAS 100, 5268–5273 (2003), Nachman et al: "The genetic basis of adaptive melanism in pocket mice". © 2003 by The National Academy of Sciences

If we know the adaptive phenotype, then we can map the genotype.

But, what if we don't know the phenotype? What if we only have genotypes?

How do we know which mutations are adaptive, deleterious, and neutral?

```
Human              accacagcatttgttagttactgccaagaagcctgtatctgtagggtaaaatcctcgctgaagtgggttg
Chimp              ......................................g...........c...................
Gorilla            ...................................................cc..................
Orangutan          .........c.........c...............................c...................
Gibbon             ...................c...............................---.................
Crab-eating macaque g............gg...c...............................c..t.t..............
```

How do we know which mutations are adaptive, deleterious, and neutral?

```
Human              accacagcatttgttagttactgccaagaagcctgtatctgtagggtaaaatcctcgctgaagtgggttg
Chimp              ......................................g...........c...................
Gorilla            ...................................................cc..................
Orangutan          .........c.........c...............................c...................
Gibbon             ...................c...............................---.................
Crab-eating macaque g............gg...c...............................c..t.t..............
```

E.g.,

Is this mutation adaptive because we see it in multiple taxa?

Or, was this a random fixation?

What are **neutrality** and **selection** in population genetics?

Neutral sites → randomly generated alleles, randomly assorting generation-to generation

Selected sites → alleles that produce some phenotypic change that produces a <u>non-random</u> effect/skew on those alleles frequencies in the population

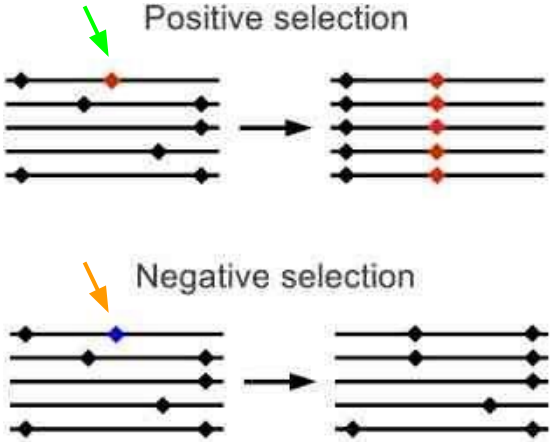
- Advantageous mutations
- Deleterious mutations

There are multiple types of (natural) **selection**

Neutral sites → randomly generated alleles, randomly assorting generation-to generation

Selected sites → alleles that produce some phenotypic change that produces a non-random effect/skew on those alleles frequencies in the population

- Advantageous mutations → Selected for
- Deleterious mutations → Selected against



Modification to HW to account for natural selection

In populations undergoing selection, each genotype has a **relative fitness (*W*)**

e.g., Population with two alleles (*R* and *r*)

Relative fitness (*W*) of each genotype (*RR*, *Rr*, and *rr*):

W_{RR} W_{Rr} W_{rr}

Relative frequencies of each genotype at adulthood (from parents to the next generation):

p^2W_{RR} $2pqW_{Rr}$ q^2W_{rr}

***Think of W as a genotype scaling factor*

Modification to HW to account for natural selection

In populations undergoing selection, each genotype has a **relative fitness (*W*)**

e.g., Population with two alleles (*R* and *r*)

Relative fitness (*W*) of each genotype (*RR*, *Rr*, and *rr*):

W_{RR} W_{Rr} W_{rr}

Relative frequencies of each genotype at adulthood (from parents to the next generation):

p^2W_{RR} $2pqW_{Rr}$ q^2W_{rr}

E.g., Frequency of q homozygotes (q^2) scaled by their relative fitness (W_{rr})

***Think of W as a genotype scaling factor*

Modification to HW to account for natural selection

In populations undergoing selection, each genotype has a **relative fitness (*W*)**

e.g., Po

Rel

Rel

ger

to the next

Recap from lecture 20:

1. Calculate sample allele frequencies from genotype frequencies:
   p = frequency (R) = frequency of RR + (1/2) frequency of Rr

   q = frequency (r) = frequency of rr + (1/2) frequency of Rr

2. Calculate HWE genotype frequencies from allele frequencies:
   $p^2 + 2pq + q^2 = 1$

p^2W_{RR} $2pqW_{Rr}$ q^2W_{rr}

E.g., Frequency of q homozygotes (q^2) scaled by their relative fitness (W_{rr})

***Think of W as a genotype scaling factor*

Fitness-modified HWE

Normalization factor, the sum of the relative contributions of each genotype to the next generation:

$$W = p^2 W_{RR} + 2pqW_{Rr} + q^2 W_{rr}$$

relative measurement of reproduction from one generation to the next

Calculating changes in allele frequencies caused by selection

To calculate the genotype frequencies <u>after selection</u>,

1) Find the zygote HWE genotype frequencies formed through random mating
2) Normalize these terms with each genotypes' normalized fitness (W)
3) Calculate the overall normalization factor $\hat{W}$

Parental Gametes

Allele	R	r
Frequency	p	q

First Generation Offspring

Genotypes	RR	Rr	rr
Zygote frequency	p^2	$2pq$	q^2
Relative fitness	W_{RR}	W_{Rr}	W_{rr}
Relative frequency after selection	p^2W_{RR}	$2pqW_{Rr}$	q^2W_{rr}
Normalized adult frequency	$\frac{p^2W_{RR}}{\overline{W}}$	$\frac{2pqW_{Rr}}{\overline{W}}$	$\frac{q^2W_{rr}}{\overline{W}}$

Gametes from First Generation

Allele	R	r
Frequency	$p' = \frac{(p^2W_{RR} + pqW_{Rr})}{\overline{W}}$	$q' = \frac{(q^2W_{rr} + pqW_{Rr})}{\overline{W}}$

Normalization factor $\overline{W} = p^2W_{RR} + 2pqW_{Rr} + q^2W_{rr}$

4) Calculate the selection-adjusted HWE allele frequencies (p' and q') produced by the offspring generation

Calculating the changes in allele frequencies due to selection

***p′* and *q′* represent allele frequencies after one generation of selection**

Copyright © McGraw-Hill Education. Permission required for reproduction or display.

$$q' = \frac{q^2 W_{aa} + \frac{1}{2}(2pqW_{Aa})}{\overline{W}} = \frac{q(qW_{aa} + pW_{Aa})}{\overline{W}} \qquad (20.5)$$

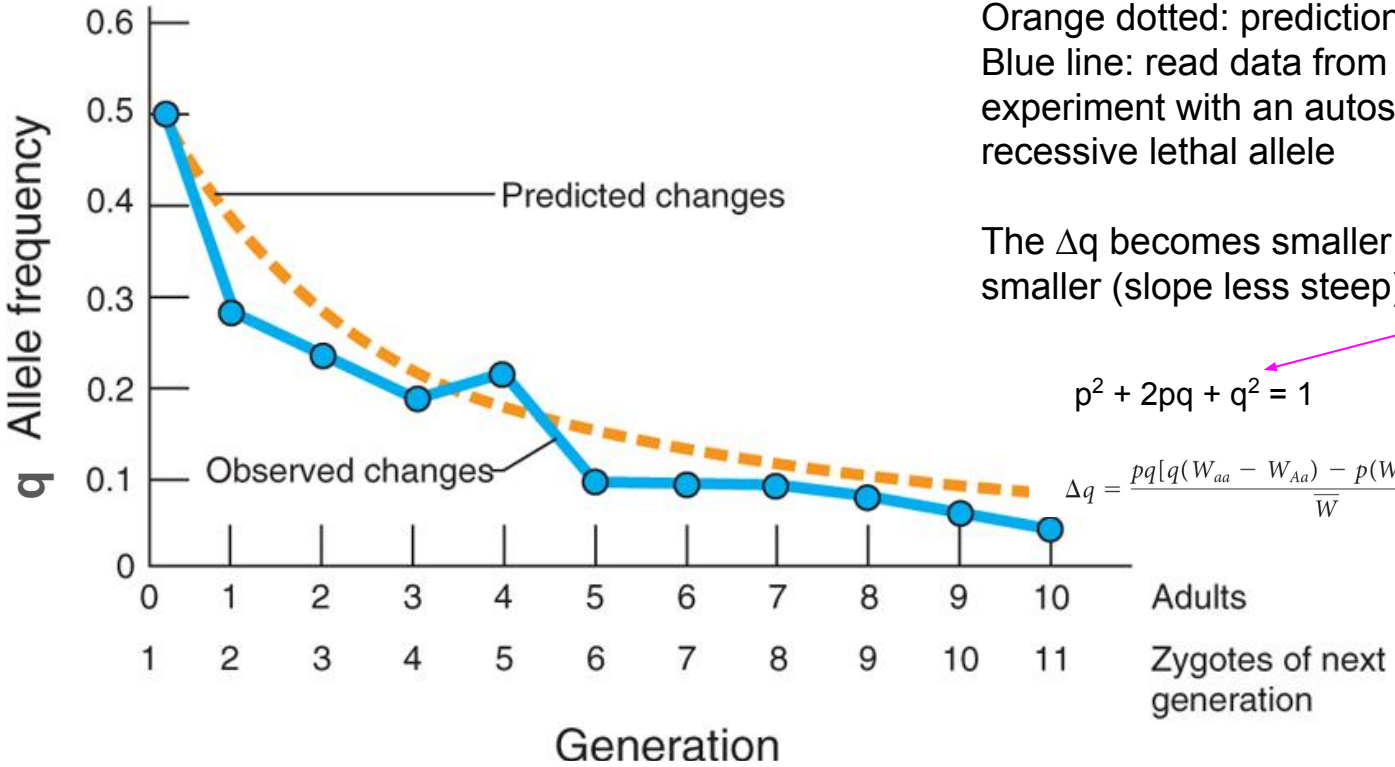
$$\Delta q = \frac{pq[q(W_{aa} - W_{Aa}) - p(W_{AA} - W_{Aa})]}{\overline{W}} \qquad (20.6)$$

Rate of decrease depends on the allele frequencies

As *q* approaches 0, rate of decrease gets slower

What is Δq if there is no selection (i.e. all relative fitness levels are equal)?

Decrease in the frequency of a recessive lethal allele over time



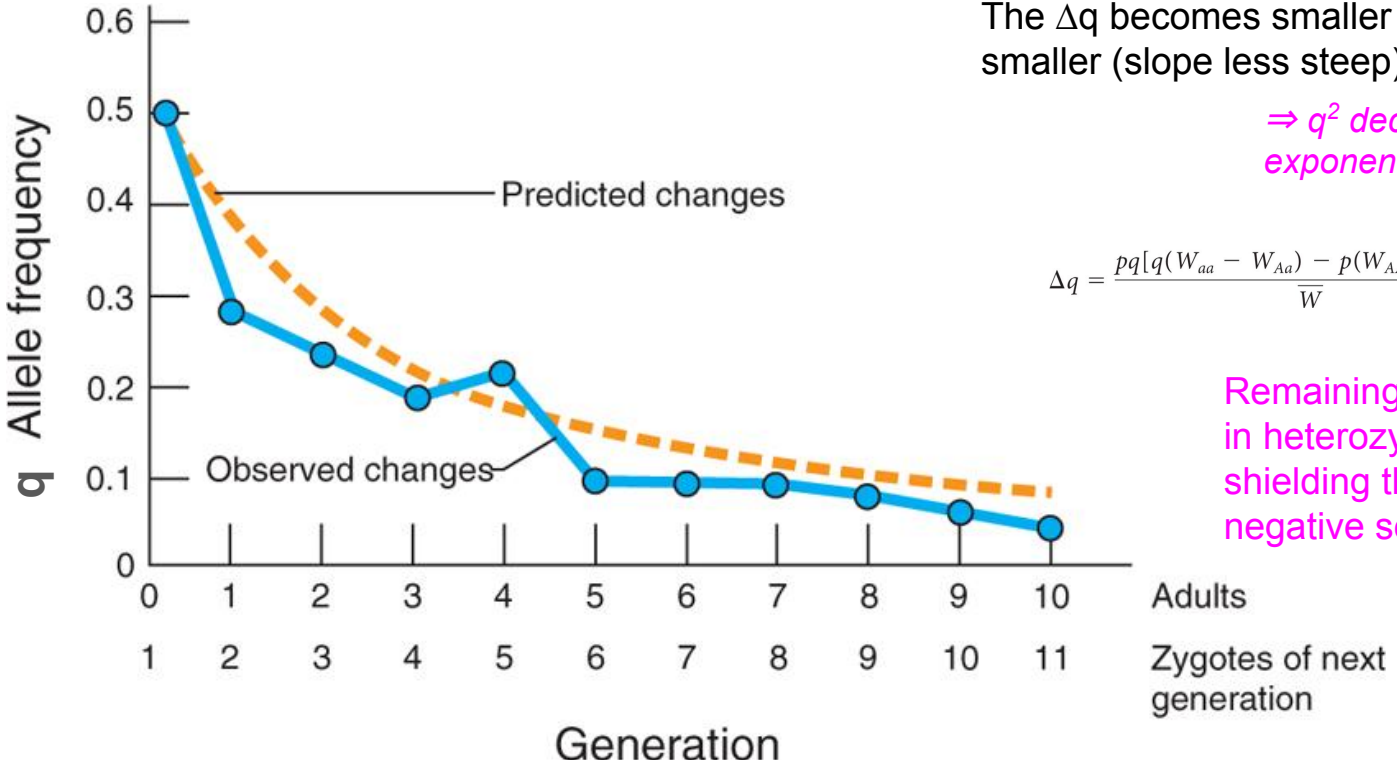
Orange dotted: prediction
Blue line: read data from an experiment with an autosomal recessive lethal allele

The Δq becomes smaller as q is smaller (slope less steep). Why?

$p^2 + 2pq + q^2 = 1$

$$\Delta q = \frac{pq[q(W_{aa} - W_{Aa}) - p(W_{AA} - W_{Aa})]}{\overline{W}} \quad (20.6)$$

Decrease in the frequency of a recessive lethal allele over time



The Δq becomes smaller as q is smaller (slope less steep). Why?

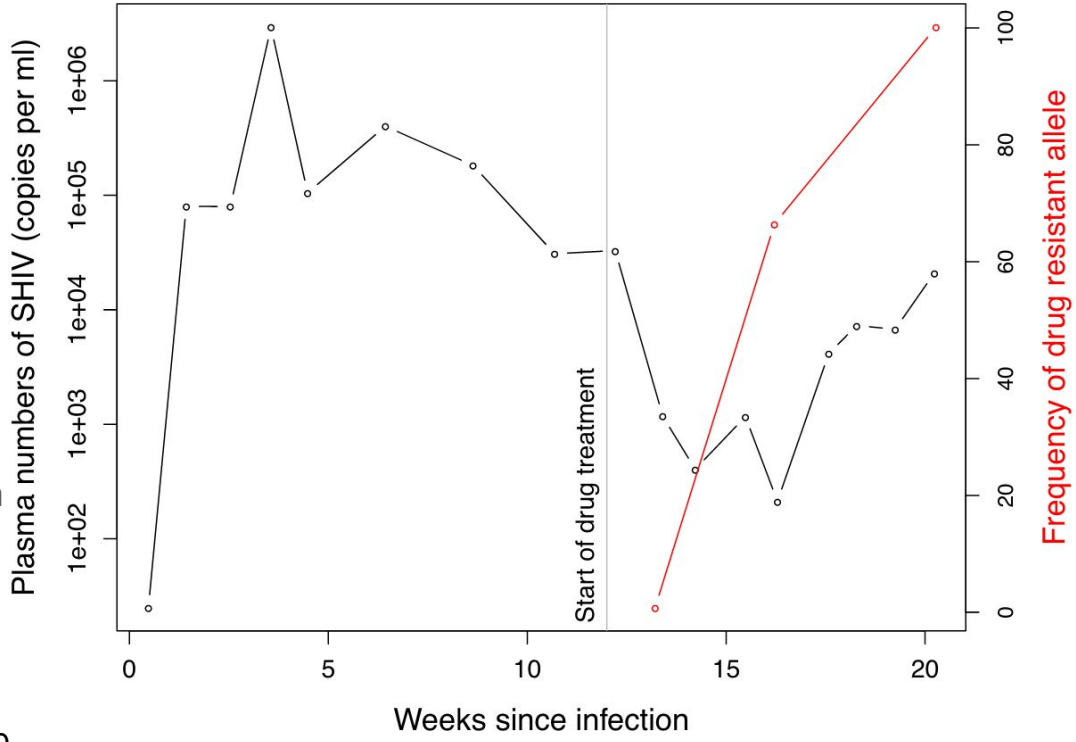
$\Rightarrow q^2$ decreases exponentially

$$\Delta q = \frac{pq[q(W_{aa} - W_{Aa}) - p(W_{AA} - W_{Aa})]}{\overline{W}} \quad (20.6)$$

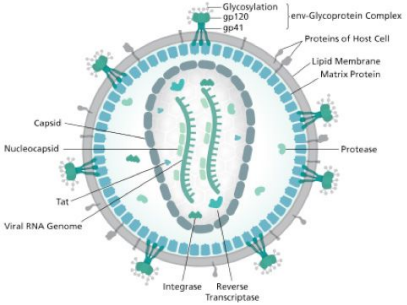
Remaining q are mostly in heterozygotes, shielding them from negative selection.

Increase in the frequency of a drug resistance mutation

The viral load of SHIV in the blood of a macaque



Simian-Human Immunodeficiency Virus

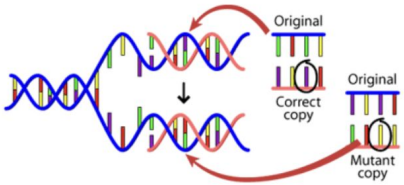


from Coop 2020 Chapter 10

How do beneficial mutations get to high frequency?

Review from Lecture 10:

Multiple alleles exist for many traits



- A gene can have multiple alleles, but each individual can only carry two alleles
- Allele frequency = copies of allele / 2x(diploid population size)
  - Wild-type alleles occur >1% of the population
  - Mutant alleles occur in <1% of the population
  - An allele can change frequency and switch categories
    - (more on this in lectures 20-21)
- Dominance relationships are always relative to a second allele and are unique to a pair of alleles

E.g., Multiple sequence alignment of blood group alleles

	Exon 3			Exon 5			Intron 6																
		Exon 4			Exon 6													Exon 7					
	106	188	189	220	261delG	297	163	179	271	280	446	628	786	891	901	950	526	657	703	796	803	930
A^1	G	G	C	C	G	A	T	C	A	C	A	A	A	A	G	A	C	C	G	C	G	G
B	G	G	C	C	G	G	C	T	G	T	G	G	G	G	A	G	G	T	A	A	C	A
O^1	G	G	C	C	261delG	A	T	C	A	C	A	A	A	A	G	A	C	C	G	C	G	G
O^{24} (O1v-B)	T	A	T	T	261delG	G	/	/	/	/	/	/	/	/	/	/	G	T	A	A	C	A
O^{41} (O1v-B, Ile513)	/	/	/	/	261delG	G	C	T	G	T	G	G	G	G	A	G	G	T	A	A	C	A

How do beneficial mutations get to high frequency?

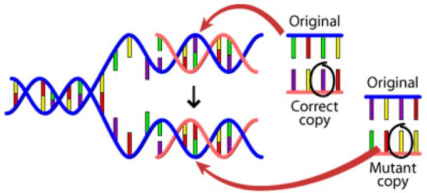
In diploids, heterozygotes hide recessive alleles from selection
→ slows both fixation and loss

If selection pressures are strong, alleles can rise in frequency by *W* alone

Weak selection can influence allele frequency, but it must overcome genetic drift

How do beneficial mutations get to high frequency?

mutation → substitution/allele → selected allele



change in DNA sequence

DNA variant present in multiple individuals ~ also called a segregating site [in the population]

DNA variant encoding a phenotype with differential impacts on survival and/or fitness

Natural selection cannot overcome small population size

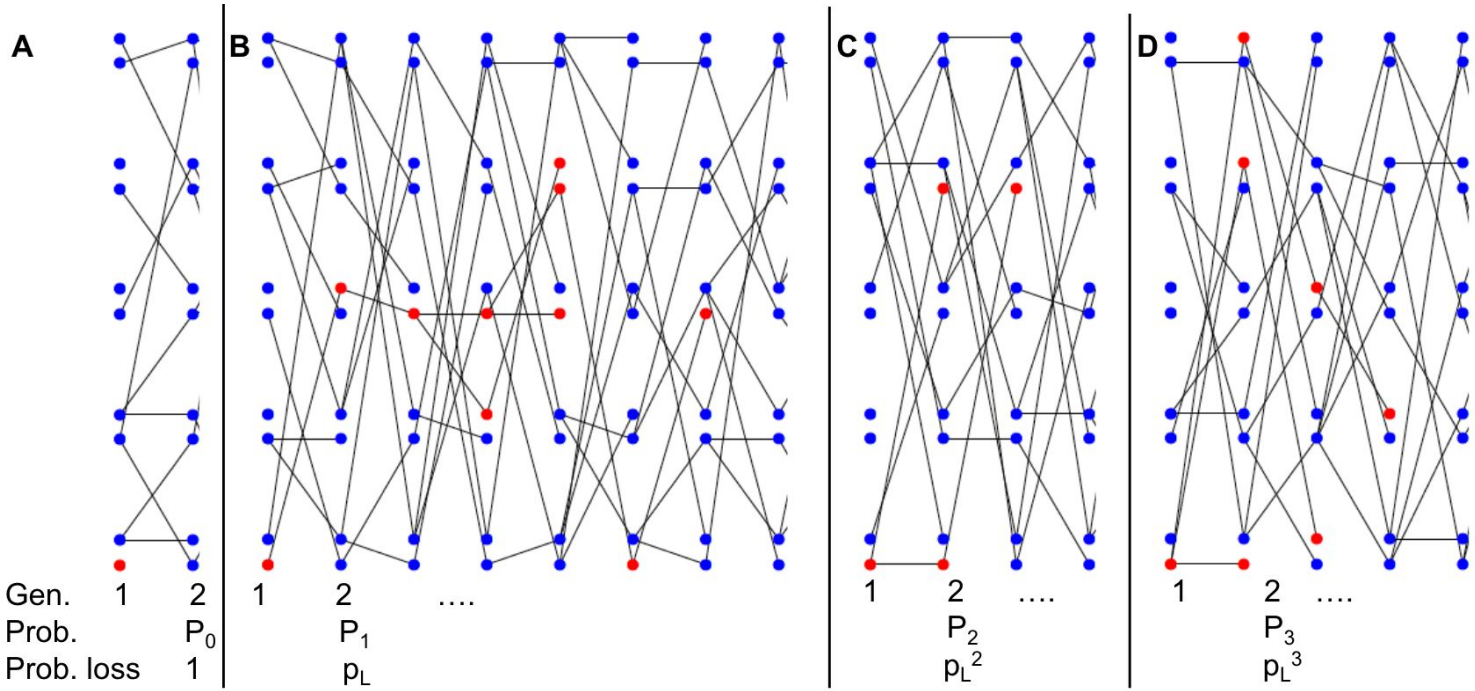


Figure 12.1: Four different outcomes of a selected allele present as a single copy in the population, leaving zero, one, two, three offspring in the next generation.

from Coop 2020 Chapter 12

Larger population sizes make selection more effective & approach theoretical expectations

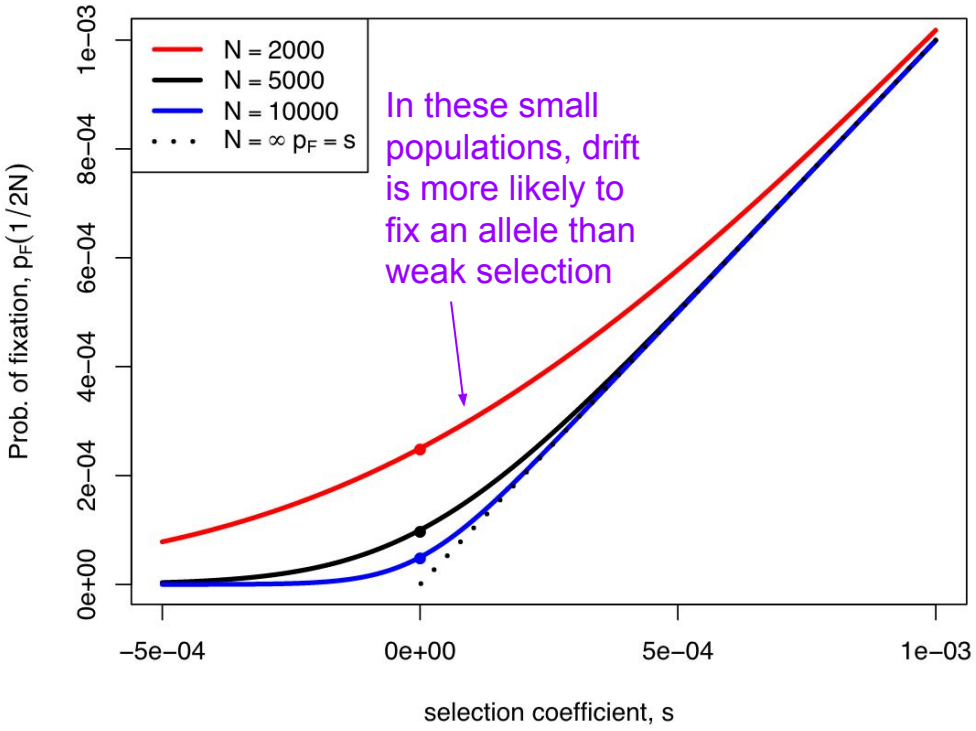


Figure 12.7: The probability of the fixation of a new mutation with selection coefficient s ($h = 1/2$) in a diploid population of effective size N_e. The dashed line gives the infinite population solution. The dots give the solution for $s \to 0$, i.e. the neutral case, where the probability of fixation is $1/2N_e$. Code here.

Ne = idealized population size, if all assumptions are correct

Note: selection coefficient (s) ~ relative selection coefficient W_{aa} (heterozygotes are scaled by the dominance coefficient, h)

from Coop 2020 Chapter 12

The efficacy of selection depends on population size

Only strongly selected mutations are likely to fix in small populations

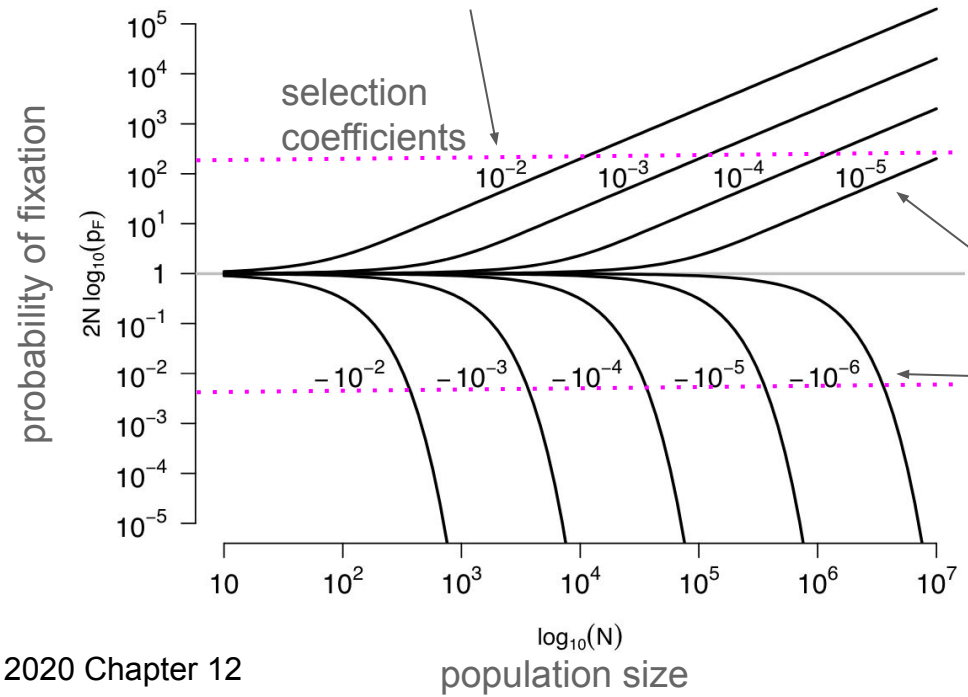


Figure 12.8: The probability of the fixation of a new mutation with selection coefficient s relative to the neutral fixation probability ($1/2N$) as a function of the effective size N. The selection coefficient is shown next to eacj line, deleterious alleles have negative selection coefficients. Note how quickly the probabilities move away from the neutral expectation as Ns moves past 1. Code here.

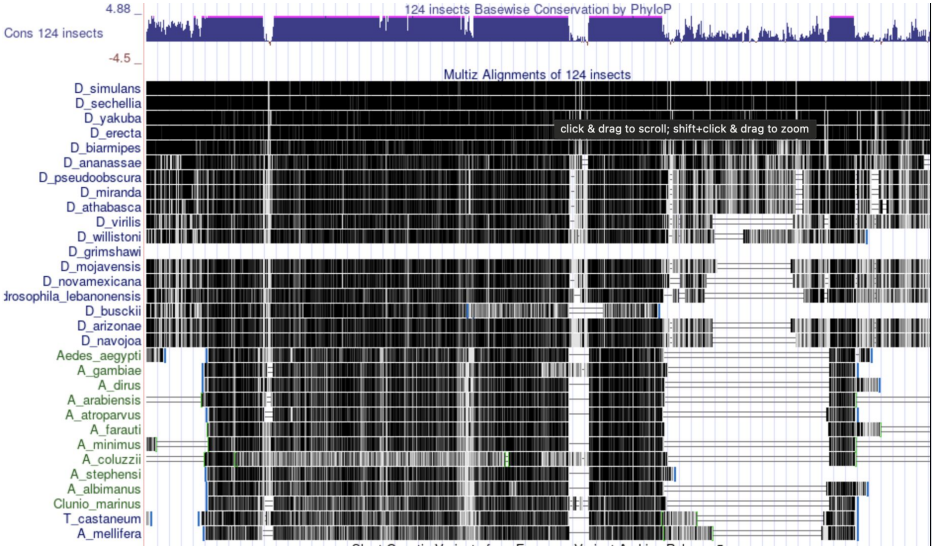
Weakly selected mutations are unlikely to fix in small populations ~ they are only efficiently selected in large populations

Note: selection coefficient (s) ~ relative selection coefficient W_{aa} (heterozygotes are scaled by the dominance coefficient, h)

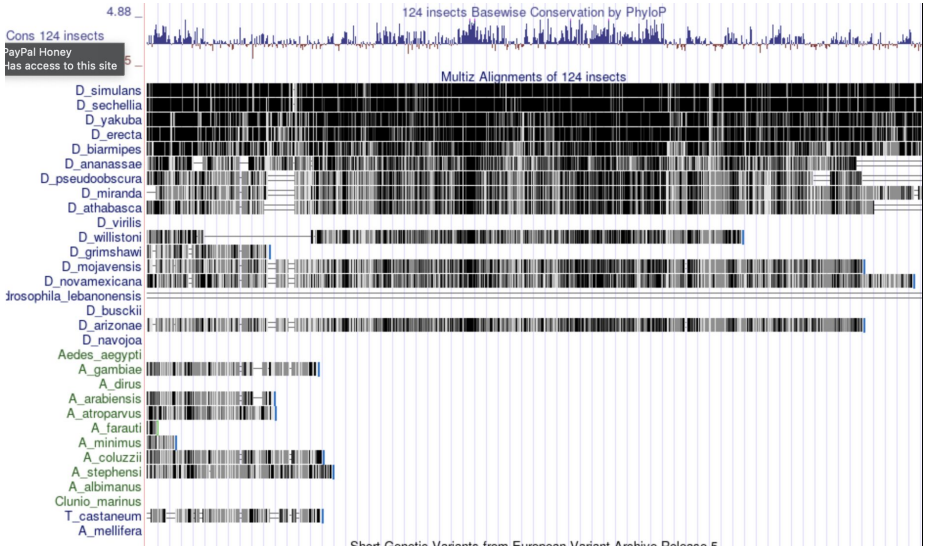
from Coop 2020 Chapter 12

Selection against substitutions results in long-term molecular conservation (to retain function)

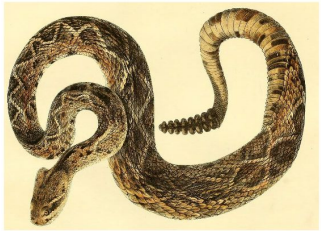
Kinesin heavy chain motor protein from insects is essential for intracellular microtubule motility and is highly conserved



Bag of marbles protein from insects is occasionally essential for male or female germline development, and evolves quickly



The absence of substitutions indicates constraint



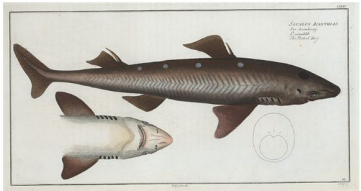
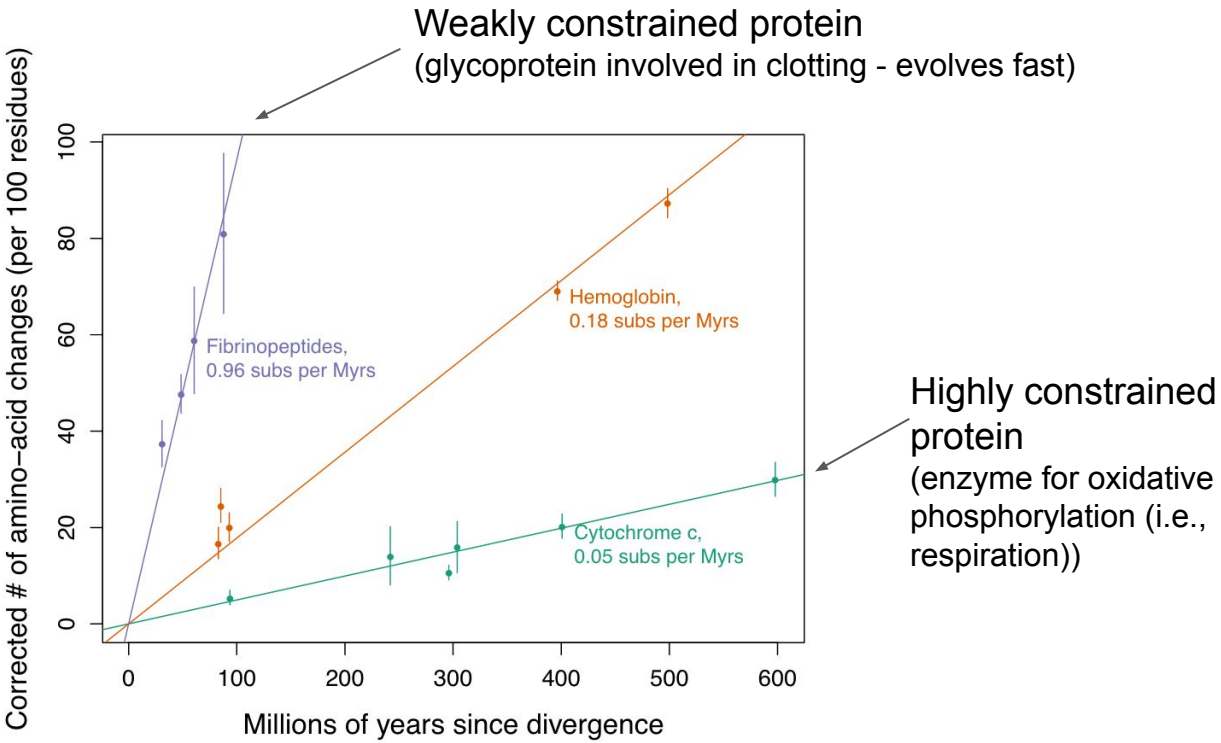


Figure 5.3: The numbers of substitutions in three proteins, corrected for multiple hits, between various pairs of groups plotted against the time these groups shared a common ancestor in the fossil record. Data from DICKERSON (1971). The lines give the linear regression through the origin for each protein. The slope of the regression is given next to the protein name. Code here. See (ROBINSON *et al.*, 2016) who revisited this classic study and confirmed the conclusions.



Coop 2020

Genetic diversity is maintained despite constraint (stabilizing selection) because of weak and balancing evolutionary forces

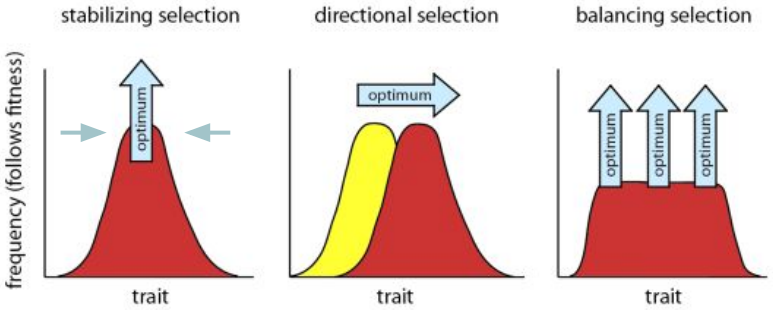
Stabilizing selection
→ optimize or retain function (resulting in molecular constraint/conservation)

New mutations
→ regularly created and lost or fixed at mutation-selection equilibrium

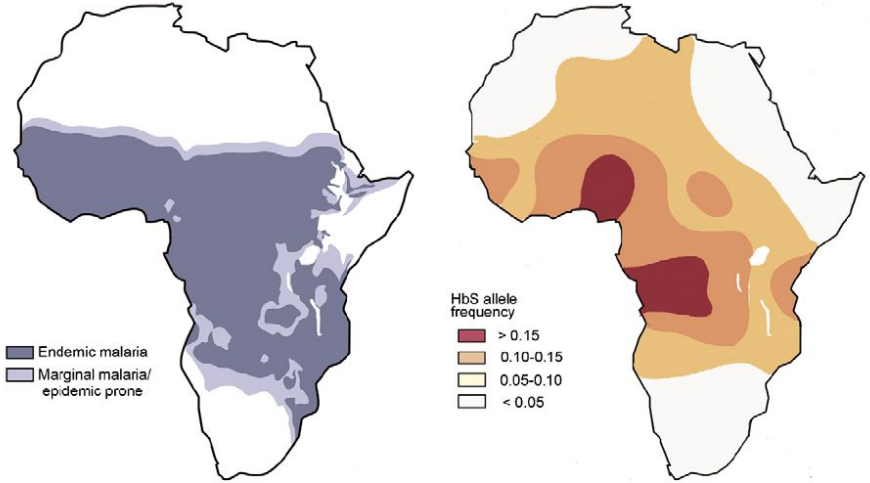
Weak directional selection
→ deleterious mutations fail to be lost and advantageous mutations fail to be fixed

Masking of recessive alleles in heterozygotes
→ slow selection of alleles with phenotypes only in homozygotes

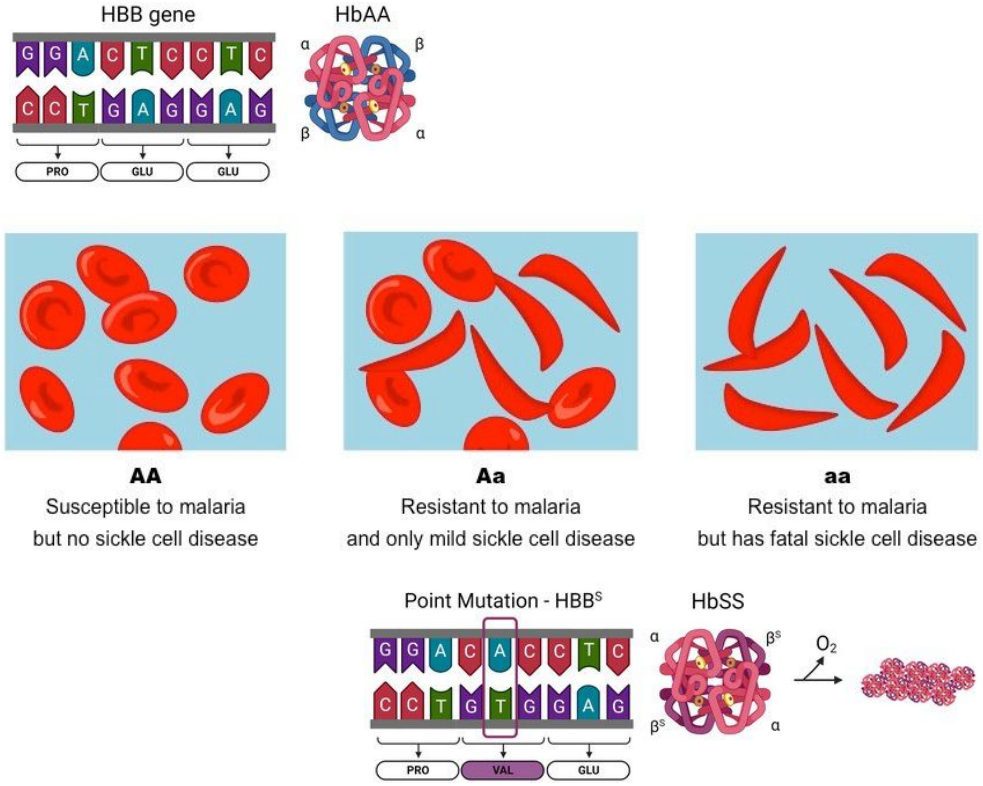
Balancing selection
→ advantage to maintaining multiple alleles



Balancing selection example: heterozygote advantage in sickle cell anemia



Williams 2006 Current Opinion in Microbiology



Ramadas and Sparkenbaugh 2023 Frontiers in Medicine

How do we find mutations under selection?

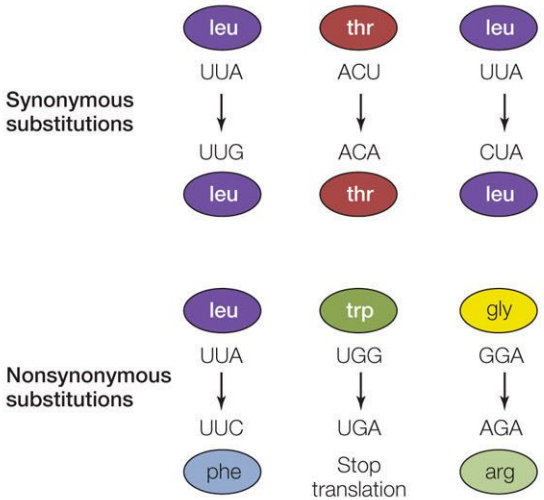
There are many population genetic signatures of selection…
we'll cover one involving coding genes in this class

Detecting signatures of selection

How would we find “fitter” genotypes in certain populations?

Measure the relative rate of non-synonymous amino acid substitutions

$$\frac{d_n}{d_s}$$ = number of non-synonymous mutations / synonymous mutations



1st letter	Second Letter: U		Second Letter: C		Second Letter: A		Second Letter: G		3rd letter
U	UUU	Phe	UCU	Ser	UAU	Tyr	UGU	Cys	U
	UUC		UCC		UAC		UGC		C
	UUA	Leu	UCA		UAA	Stop	UGA	Stop	A
	UUG		UCG		UAG	Stop	UGG	Trp	G
C	CUU	Leu	CCU	Pro	CAU	His	CGU	Arg	U
	CUC		CCC		CAC		CGC		C
	CUA		CCA		CAA	Gln	CGA		A
	CUG		CCG		CAG		CGG		G
A	AUU	Ile	ACU	Thr	AAU	Asn	AGU	Ser	U
	AUC		ACC		AAC		AGC		C
	AUA		ACA		AAA	Lys	AGA	Arg	A
	AUG	Met	ACG		AAG		AGG		G
G	GUU	Val	GCU	Ala	GAU	Asp	GGU	Gly	U
	GUC		GCC		GAC		GGC		C
	GUA		GCA		GAA	Glu	GGA		A
	GUG		GCG		GAG		GGG		G

Degeneracy of 3rd position allows for synonymous mutations

= nucleotide mutations that do not change the amino acid

Detecting signatures of selection

How would we find “fitter” genotypes in certain populations?

Measure the relative rate of non-synonymous amino acid substitution $\mathbf{d_n/d_s}$

$\frac{\mathbf{d_n}}{\mathbf{d_s}}$ = 1 → no selection (no pressure or ineffective) ← Pseudogenes (loss of function)

< 1 → negative or purifying selection (*i.e.*, conservation) ← Most protein-coding genes (in most genomes) ~ constraint

> 1 → positive selection (*i.e.,* adaptation)

Some genes are variably constrained, illustrating the power of selection over evolutionary time

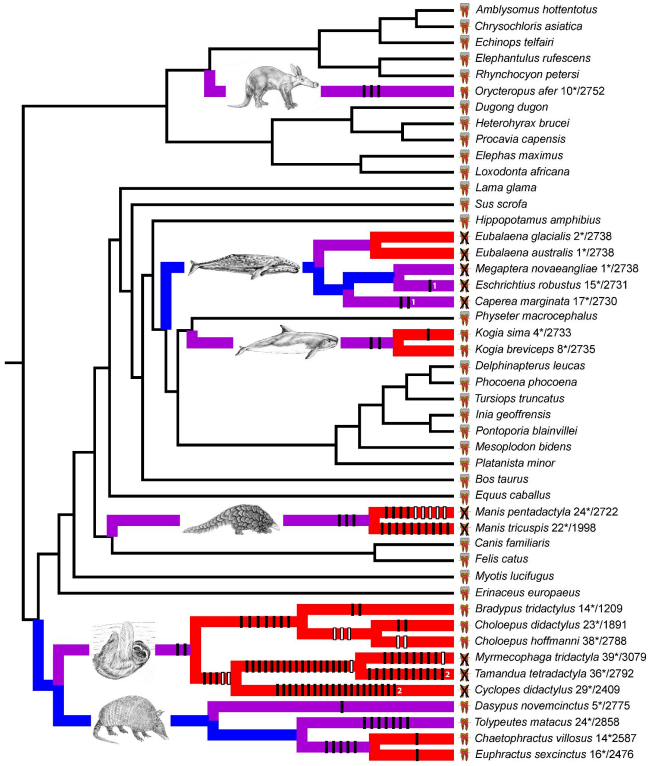


Figure 1. Species tree with frameshift mutations and dN/dS branch coding. Symbols next to taxon names denote taxa having teeth with enamel, taxa having teeth without enamel, and edentulous taxa. Branches are functional (black), pre-mutation (blue), mixed (purple), and pseudogenic (red). Vertical bars on branches represent frameshift mutations (see Table S1). Frameshifts that map unambiguously onto branches are shown in black. Frameshifts shown in white are unique, but occur in regions where sequences are missing for one or more taxa (Figure S7) and were arbitrarily mapped onto the youngest possible branch. Homoplastic frameshifts (deltran optimization) are marked by numbers. Numbers after taxon names indicate the minimum number of stop codons in the sequence (before slashes) and the length of the sequence (after slashes).

Loss of tooth enamel in whales, anteaters, and sloths
(due to diet relieving selection pressures)

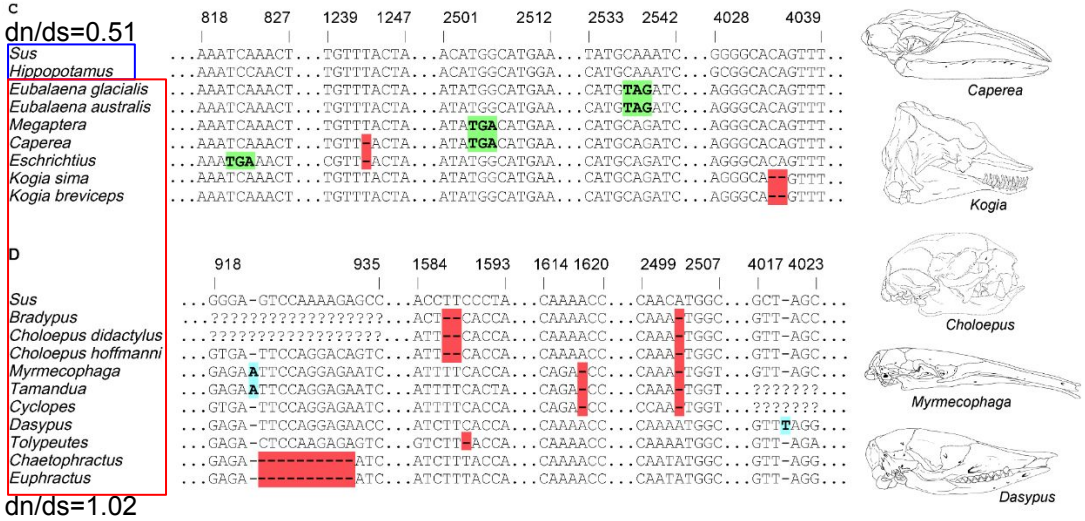


Figure 5.7: Examples of frameshift mutations (insertions blue, deletions red) and premature stop codons in *enamlin* in Cetacea and Xenarthra. Figure from Meredith *et al.* (2009), licensed under CC BY 4.0.

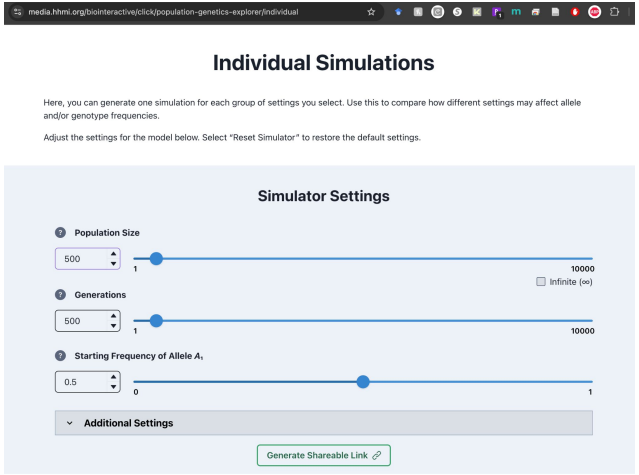
Explaining adaptation in the pocket mouse

Explaining adaptation in the pocket mouse

The Natural Selection and Adaptation film says that if dark-colored fur gives mice a 1% advantage, and you start with 1% of the population being dark-colored, it takes about 1,000 years for 95% of the mice to have dark-colored fur.

How did they come to this conclusion?

How could you test this claim?



$$\Delta q = \frac{pq[q(W_{aa} - W_{Aa}) - p(W_{AA} - W_{Aa})]}{\overline{W}} \qquad (20.6)$$

These would need to be genotyped

f(aa) = 0.01, f(aA and AA) = 0.99

Dark fur: W_{aa} = 1.01, Light fur: W_{Aa} and W_{AA} = 1

$\hat{W} = W_{aa}q^2 + W_{Aa}2pq + W_{AA}p^2$

$\Delta q = pq\,[q(1.01-1) - p(1-1)] / \hat{W}$

For next class:

Problem Set 21: Adaptation

Continue reading in Genetics: From Genes to Genomes

a. Finish Chapter 21.2
b. Start Chapter 15.1, 15.3, 15.4

Come to discussion section and office hours with questions!